

Diagnosis and Differentiation

Each of these conditions can be differentiated by appropriate porphyrin studies. Careful evaluation of urine, stool, and plasma porphyrins will almost always permit confirmation of the diagnosis of **porphyria cutanea tarda (PCT)**. Nonsteroidal anti-inflammatory drugs, such as naproxen, and antibiotics, such as tetracyclines and nalidixic acid, as well as a variety of other agents, may rarely produce bullous lesions closely resembling PCT; however, in contrast to true PCT, these cases show no evidence of abnormal porphyrins (see **pseudoporphyria**).

Treatment

General Approach

Initially, a careful history should be taken to identify potential environmental triggers (e.g., alcohol, estrogen, or chlorinated hydrocarbons) that may have induced the disease. If **HCV** or **HIV** infection is present, these should be managed appropriately. Elimination of the triggering agent alone may lead to gradual improvement. However, in most patients with PCT, more aggressive treatment is usually warranted to accelerate therapeutic benefit. Current treatment options include repeated **phlebotomy**, oral administration of **antimalarials** (chloroquine or hydroxychloroquine), or a combination of both. Other reported treatments include **iron chelators** and oral **cholestyramine**.

Phlebotomy

Phlebotomy remains the **treatment of choice** for PCT. Numerous studies have demonstrated its safety and efficacy, first introduced by Ippen. It works by depleting excessive hepatic iron stores, a hallmark of PCT. Biochemical remission has been achieved in patients both with and without overt iron overload. Replenishment of iron after remission has led to recurrence of both biochemical and clinical disease activity. Abstinence from environmental

triggers—particularly alcohol—may induce remission, though this process can take months to years.

The therapeutic efficacy of phlebotomy may be attributed to several mechanisms:

1. **Iron effects on ALA synthase:** Iron enhances the induction of **ALA synthase** in response to drugs and hepatic porphyrinogenic stimuli (e.g., hexachlorobenzene). Depleting iron reduces the inducibility of ALA synthase, thereby decreasing hepatic porphyrin production.
2. **Iron depletion effects on other heme pathway enzymes:** Ferrous iron inhibits **uroporphyrinogen (UROGEN) decarboxylase** and increases the rate of hepatic porphyrin synthesis from ALA or PBG. Iron removal may restore UROGEN decarboxylase activity and reduce excessive porphyrinogenesis. This mechanism has been supported by studies in genetically engineered mice.